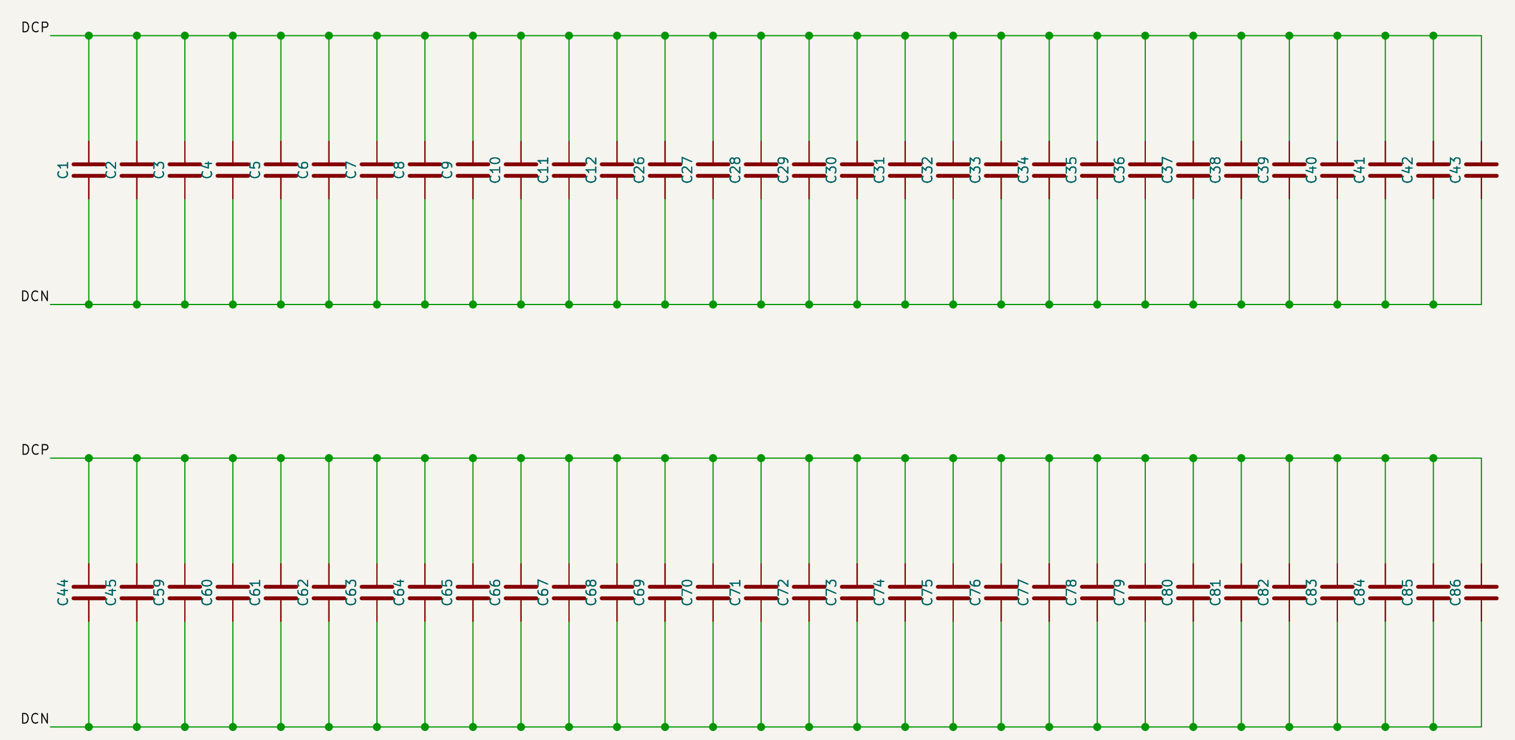
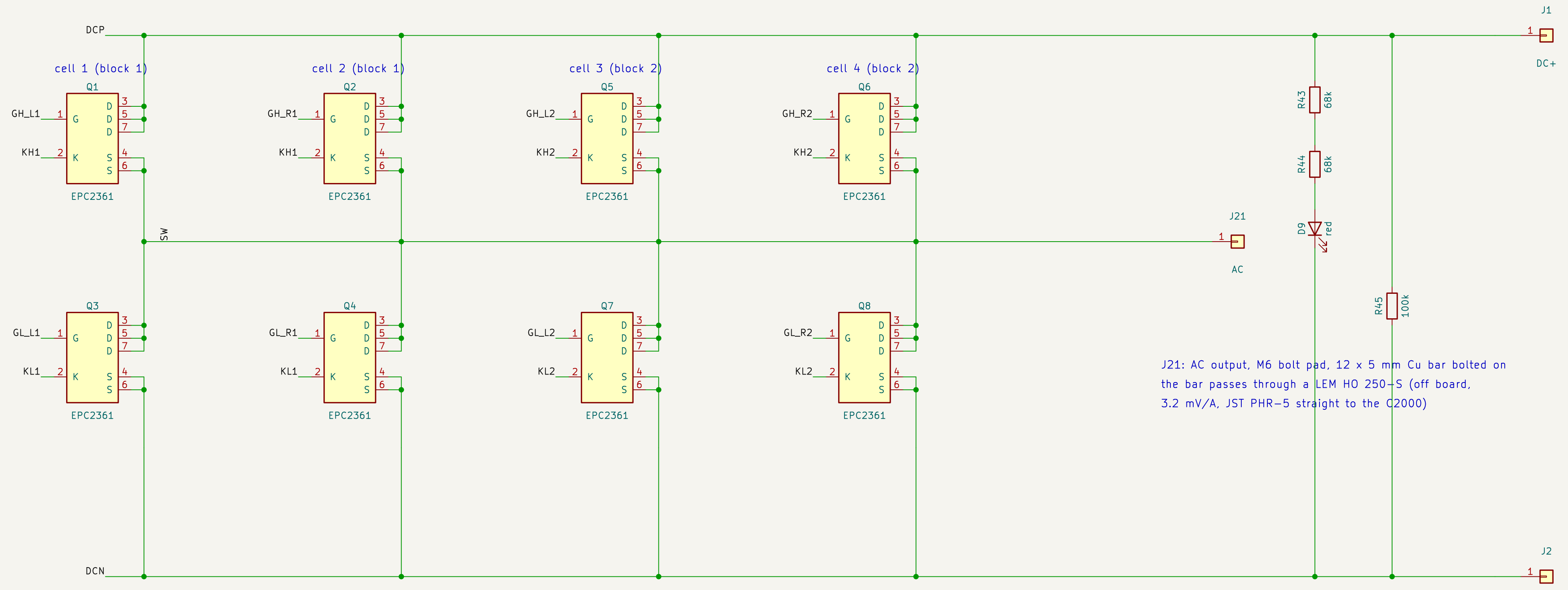
POWER STAGE AND DC LINK (block 1: Q1–Q4, block 2: Q5–Q8; one cell = HS + LS transistor, LS source straight to DC–; AC of all four cells joined on the PCB)


DCP
DCN
C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C26 C27 C28 C29 C30 C31 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41 C42 C43
C44 C45 C59 C60 C61 C62 C63 C64 C65 C66 C67 C68 C69 C70 C71 C72 C73 C74 C75 C76 C77 C78 C79 C80 C81 C82 C83 C84 C85 C86

DC link, 60 capacitors: 40 x 1uF 100V X7S, 20 x 22uF 100V X7S 2220 (1 uF 0805: bypass at the HS drains of each cell, top + bottom; 22 uF 2220: bank)



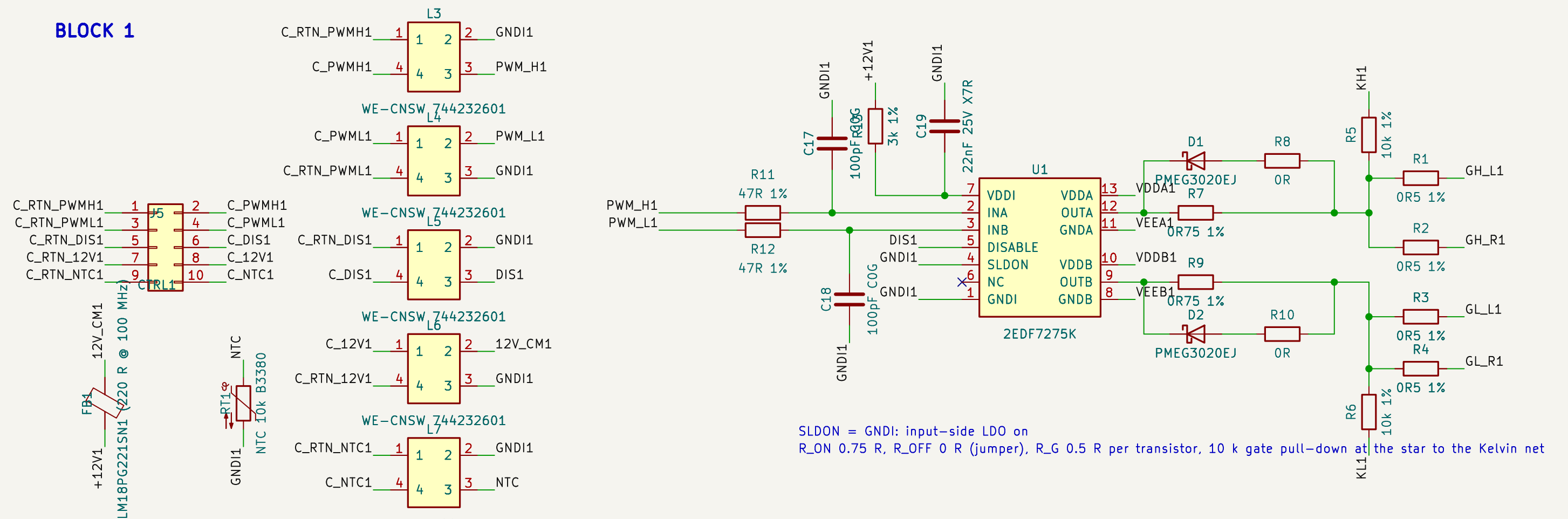
J21: AC output, M6 bolt pad, 12 x 5 mm Cu bar bolted on the bar passes through a LEM HO 250–S (off board, 3.2 mV/A, JST PHR–5 straight to the C2000)

No shunt (load–test board): LS source stripes via–in–pad (blind L1–L2) into the DC– plane

DC link present LED, R45 100 k bleeder

GATE DRIVER BLOCKS: control header with common–mode chokes, 2EDF7275K input filters, R_ON / D + R_OFF, star, R_G per transistor (block 1 drives Q1–Q4, block 2 drives Q5–Q8)

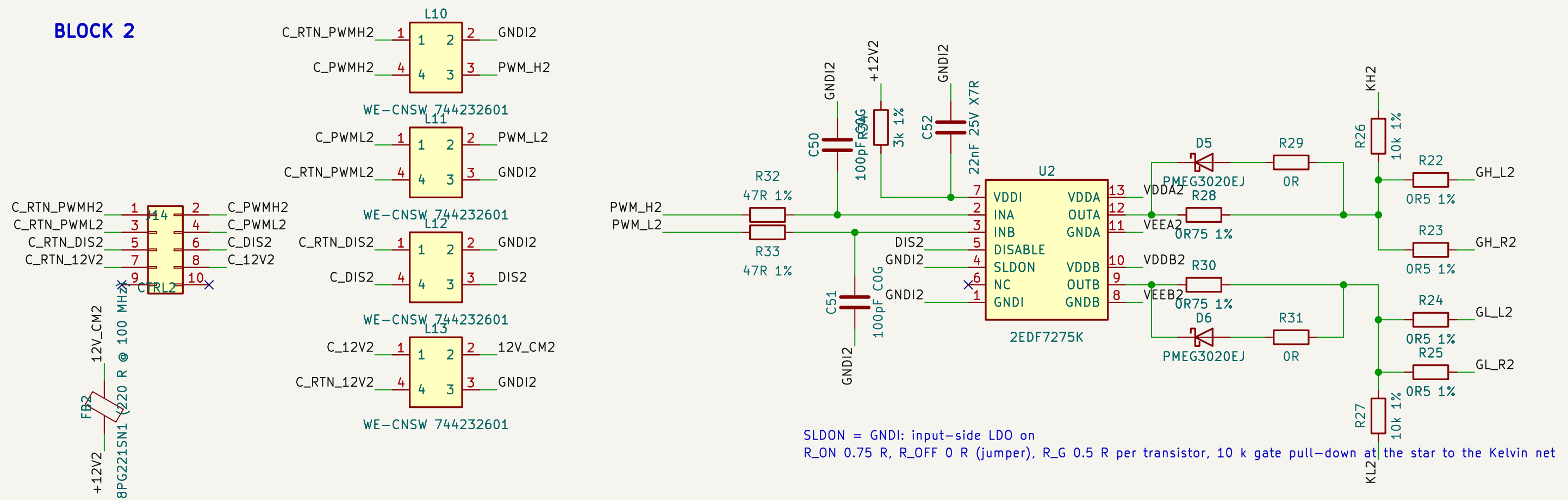
BLOCK 1



SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

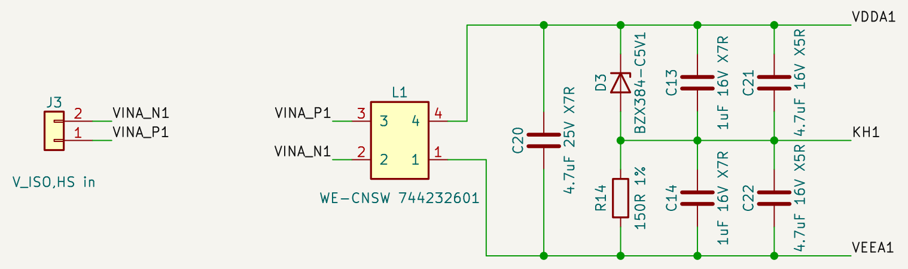
NTC on the control side (board temperature at the driver)

BLOCK 2

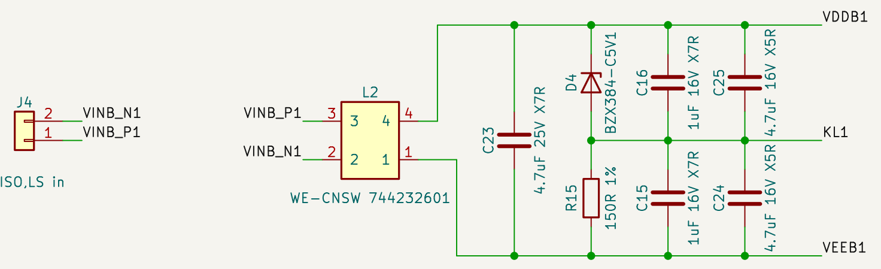


SLDON = GNDI: input–side LDO on
R_ON 0.75 R, R_OFF 0 R (jumper), R_G 0.5 R per transistor, 10 k gate pull–down at the star to the Kelvin net

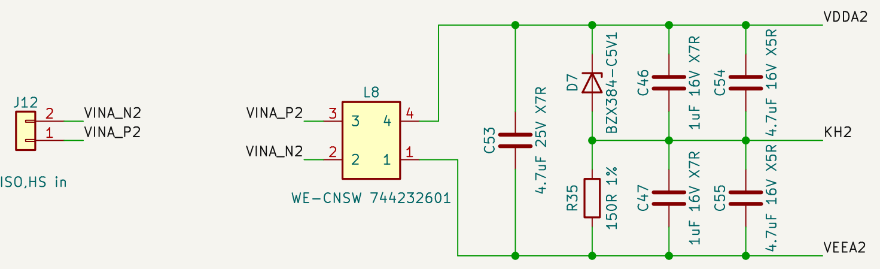
GATE SUPPLY ISLANDS: external isolated supply (one HS, one LS, split per block) –> WE–CNSW –> 5.1 V Zener VDD–S, 150 R bias S–VEE: V_GS,on = +5.1 V, V_GS,off = –(V_ISO – 5.1 V)



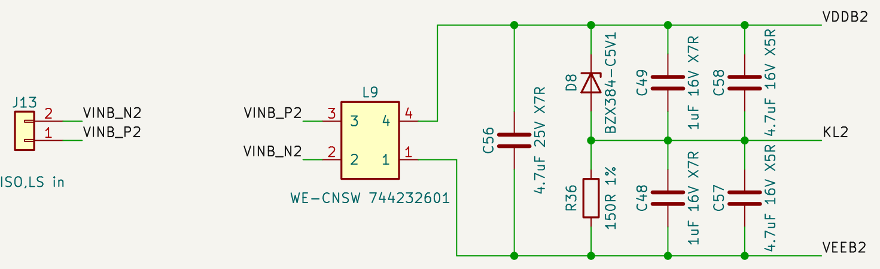
J3 / L1: HS block 1



J4 / L2: LS block 1

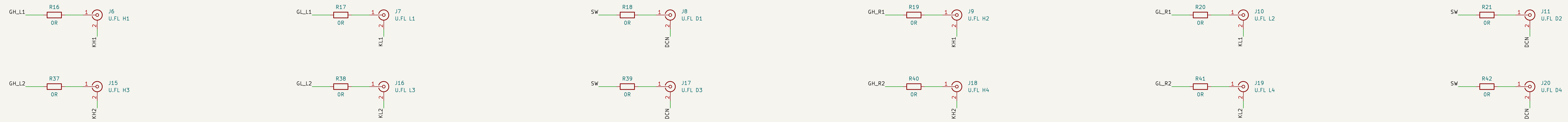


J12 / L8: HS block 2



J13 / L9: LS block 2

U.FL MEASUREMENT PORTS (bottom side, at each cell): V_GS HS / LS (0 R, shield = Kelvin), V_DS LS (4.99 k 2512 series into the 50 R scope input = 100:1, 1.1 W at 75 V; shield = LS source = DC–)


GH_L1 R16 0R J6 U.FL H1 KH1
GL_L1 R17 0R J7 U.FL L1 KL1
SW R18 0R J8 U.FL D1 DCN
GH_R1 R19 0R J9 U.FL H2 KH1
GL_R1 R20 0R J10 U.FL L2 KL1
SW R21 0R J11 U.FL D2 DCN
GH_L2 R37 0R J15 U.FL H3 KH2
GL_L2 R38 0R J16 U.FL L3 KL2
SW R39 0R J17 U.FL D3 DCN
GH_R2 R40 0R J18 U.FL H4 KH2
GL_R2 R41 0R J19 U.FL L4 KL2
SW R42 0R J20 U.FL D4 DCN

MOUNTING: H1–H4 board corners (insulated), H5–H7 heatsink spreader; heatsink HS1 (Fischer LAM 4, 100 mm, rotated 90 deg, tube across the board width) is a board–only item

Nets: DCP = DC+, DCN = DC–, SW = joined switch node of all four cells. KH1/KH2 and KL1/KL2 are the Kelvin source nets (pad 2 of each EPC2361); they meet the power source only inside the package.

LS power source = DC– directly (no shunt). The LS gate drive and the LS supply are referenced to the Kelvin net KLx, not to DC–.

Block suffix 1/2 on every driver–side net (GNDI1, VDDA2, ...). Control side (CTRLx 2 x 5 header): PWM_H, PWM_L, DIS, 12 V and (block 1) NTC each with their own return C_RTN_* through one WE–CNSW common–mode choke per pair; 12 V then through a ferrite bead. The control side is a top–layer–only area with a 1200 V barrier (4 mm outer, 3 mm L2) to all power copper; see reports/pcb–rules/PCB_rules_and_checklist.md.

No short–circuit protection on this board. Closed–loop control from a TI TMS320F28379D over the CTRL headers.

generated by scripts/make_kicad_sch_revb.py; netlist identical to the PCB (check_sch_vs_pcb.py)
4 x EPC2361 per switch, 2 x 2EDF7275K, no shunt, LAM 4 heatsink, external isolated supplies, U.FL p
KTH
Sheet: /
File: spb–bb–revb–load.kicad_sch
Title: SPB leg, Rev B, load–test board
Size: A1
Date: 2026–10–14
Rev: B
KiCad E.D.A. kicad–cli 7.0.10
Id: 1/1